Diagnostic BIOCHIPS

Electrophysiology *Reimagined*

DEEP ARRAY 128L HANDLING GUIDANCE

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1. PURPOSE

This document provides suggested guidance for operator handling of the Diagnostic Biochips Deep Array 128L probe.

2. PHILOSOPHY

Each process step is preceded by a representative image depicting how that process step should be properly executed.

3. SCOPE

This procedure is applicable to anyone intending to handle the Deep Array 128L Microelectrode array for research recordings.

4. REFERENCES

- 03-0037-02 SUB- DA128- 2L- Wrapped.PDF
- 03-0041-01 SUB- DA128L- Skirt.PDF
- 03-0044-01 Cable Assembly- GND_Ref- DA128L.PDF
- 02-0010-01 Device Label- DA128L

5. PROCEDURE

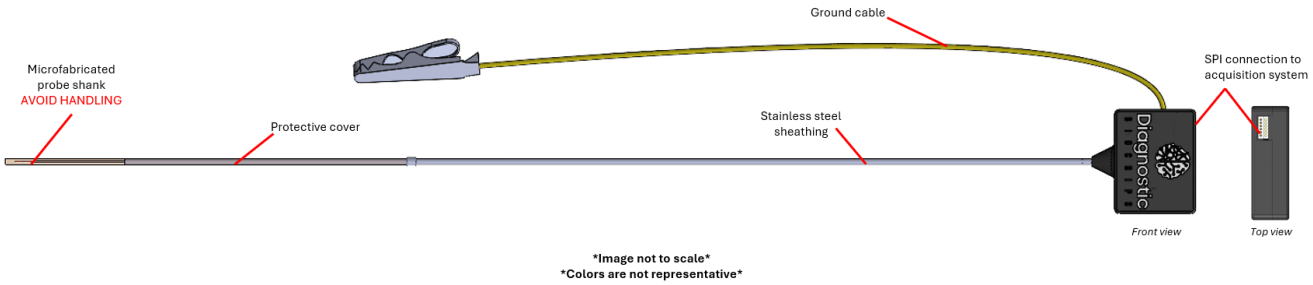
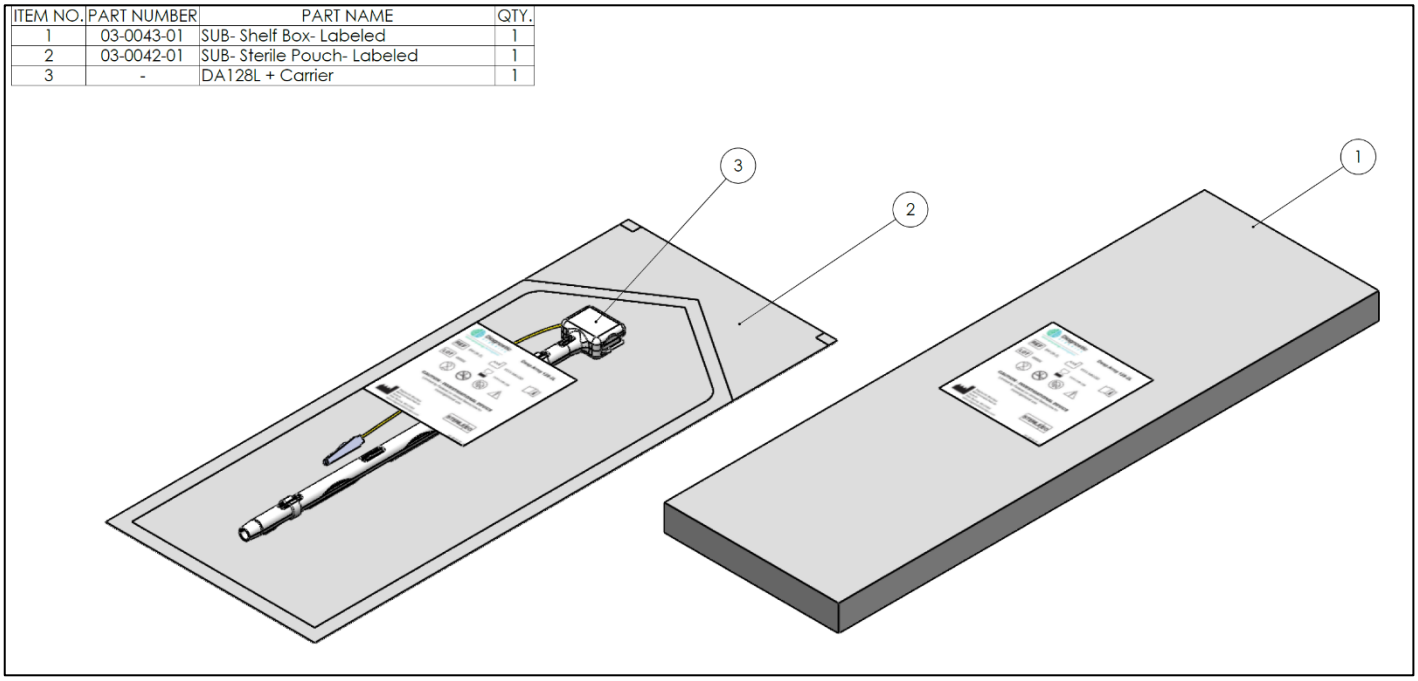


Image not to scale
Colors are not representative

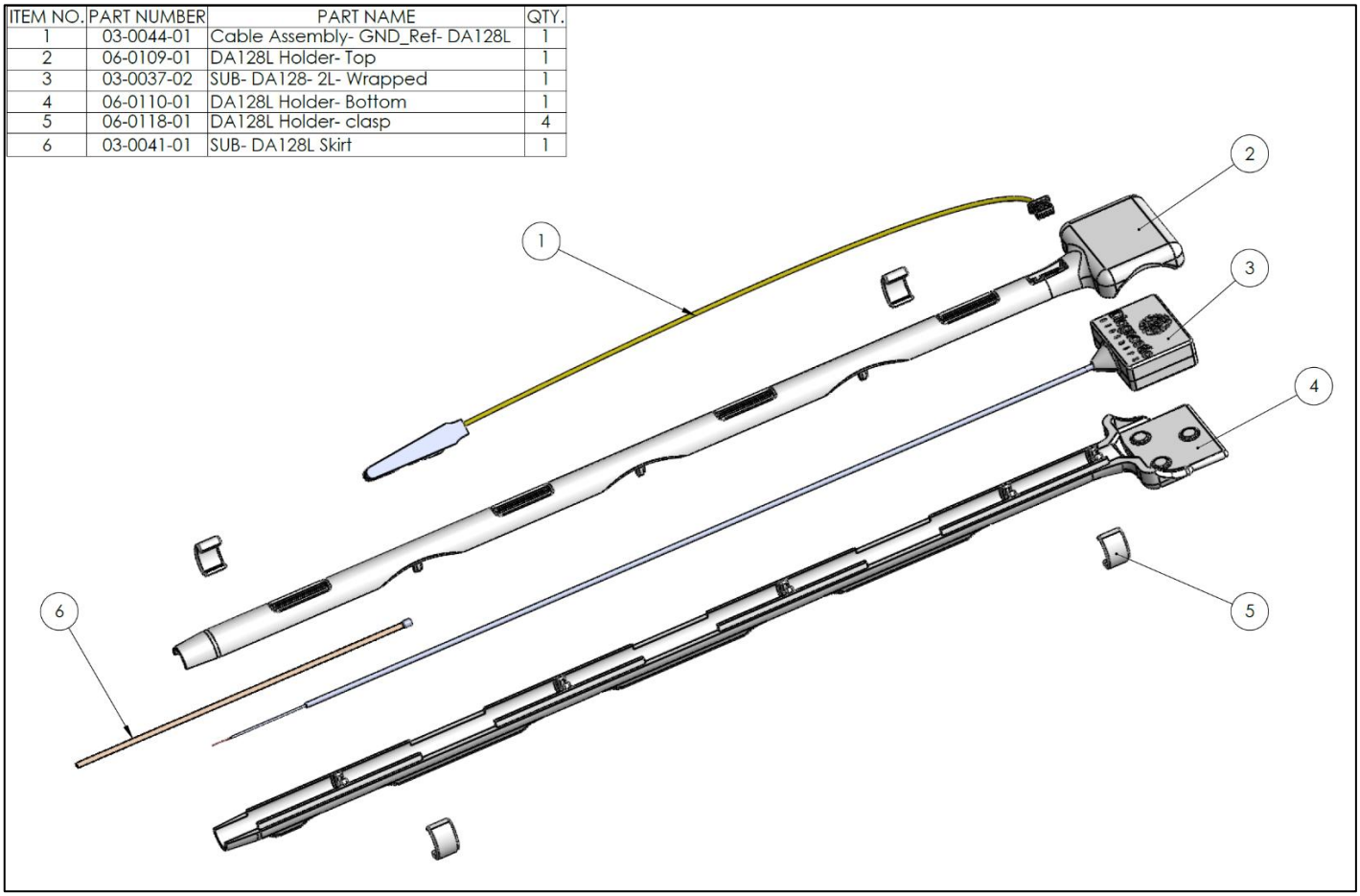
1. Research devices can be described by two separate segments, the patient contacting probe shank and the non-patient contacting backend.
    a. The patient contacting probe shank consists of a microfabricated probe tip bonded to a stainless steel microwire and reinforced by stainless steel sheathing. A protective polyimide tube covers the microfabricated probe shank until it is retracted upon insertion through a cannula.
    b. The non- patient contacting backend consists of an encapsulated Printed Circuit Board (PCB) sealed inside of a printed PCB Case and a grounding cable assembly. The stainless steel microwire and sheathing are shorted through the PCB and grounded through the cable assembly.
2. A detailed schematic of the research device is included in the attachments.
3. Research devices can be described by two separate segments, the patient contacting probe shank and the non-patient contacting backend.
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4. A detailed schematic of the research device is included in the attachments.

ITEM NO.	PART NUMBER	PART NAME	QTY.
1	03-0043-01	SUB- Shelf Box- Labeled	1
2	03-0042-01	SUB- Sterile Pouch- Labeled	1
3	-	DA128L + Carrier	1



5. Research devices will arrive with the PCB backend face up, nearest the chevron seal.

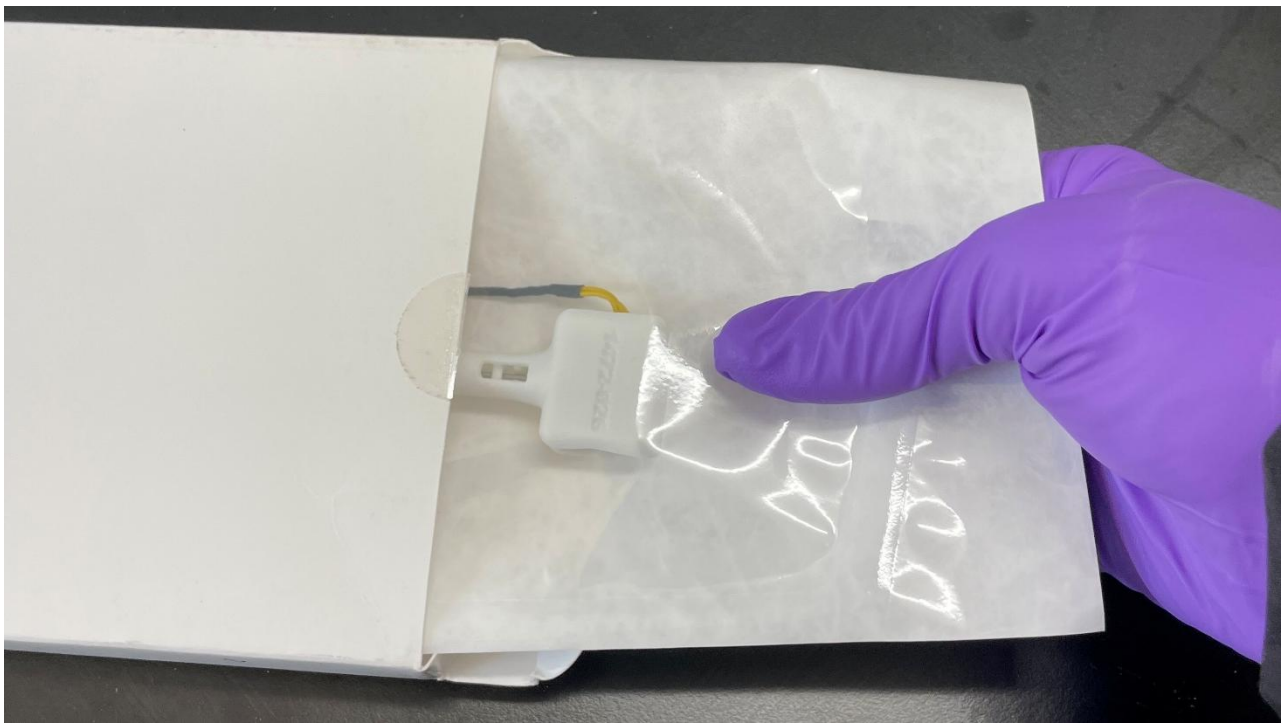
ITEM NO.	PART NUMBER	PART NAME	QTY.
1	03-0044-01	Cable Assembly- GND_Ref- DA128L	1
2	06-0109-01	DA128L Holder- Top	1
3	03-0037-02	SUB- DA128- 2L- Wrapped	1
4	06-0110-01	DA128L Holder- Bottom	1
5	06-0118-01	DA128L Holder- clasp	4
6	03-0041-01	SUB- DA128L Skirt	1



6. Inside the sterile pouch the assembly consists of 6 primary components referenced in this document and depicted above:
    a. 03-0044-01 Cable Assembly- GND_Ref- DA128L
    b. 06-0109-01 DA128L Holder- Top
    c. 03-0037-02 SUB- DA128L- 2L- Wrapped
    d. 06-0110-01 DA128L Holder- Bottom
    e. 06-0118-01 DA128L Holder- Clasp (x4)
    f. 03-0041-01 SUB- DA182L- Skirt

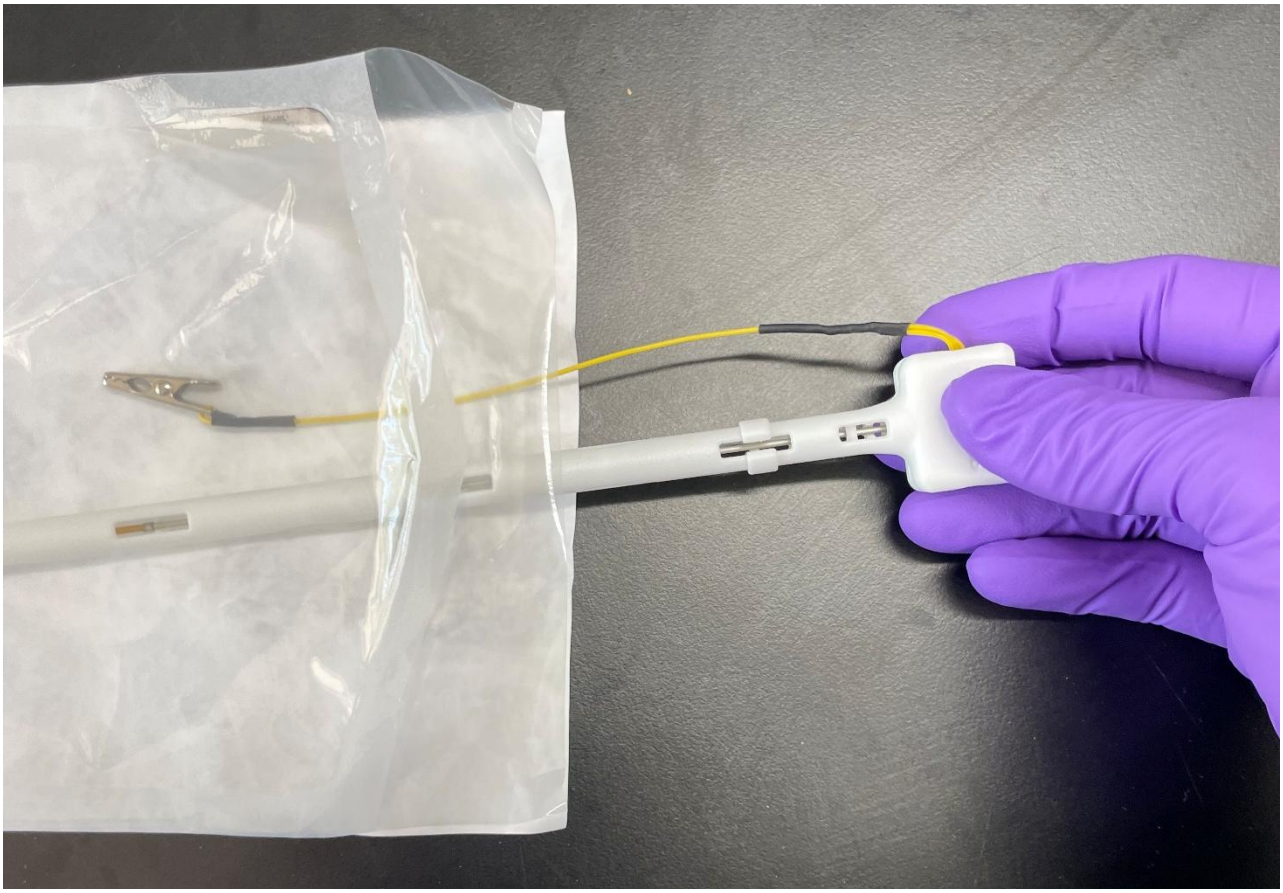


7. Ensure that the shelf box seals have not been tampered with before proceeding. Tamper seals should be applied and unbroken on both ends of the shelf box.

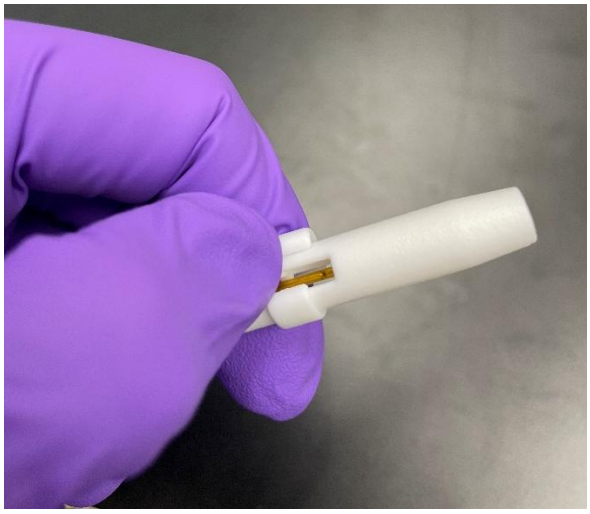


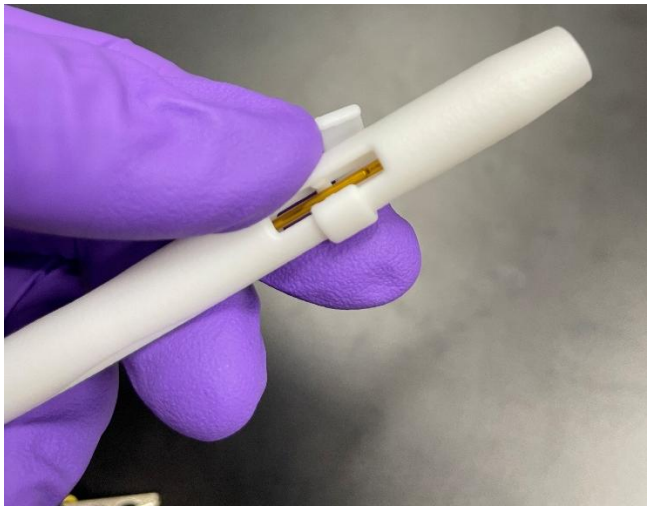
8. In a sterile environment, following sterile procedures, carefully remove the sterile pouch from the shelf box.

9. Carefully open the sterile pouch by peeling at the chevron end.

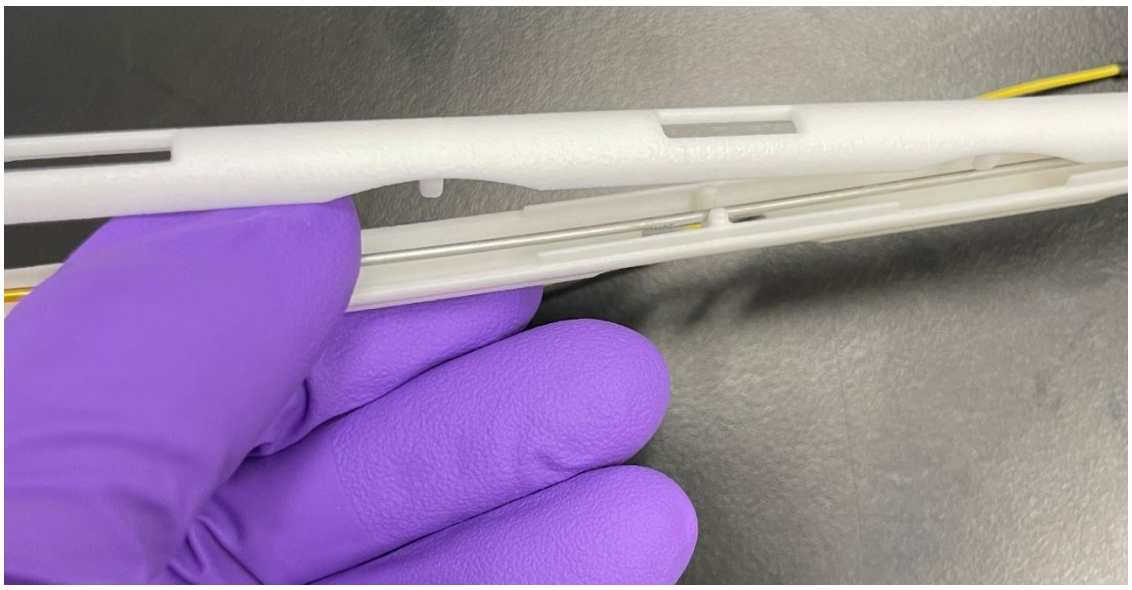


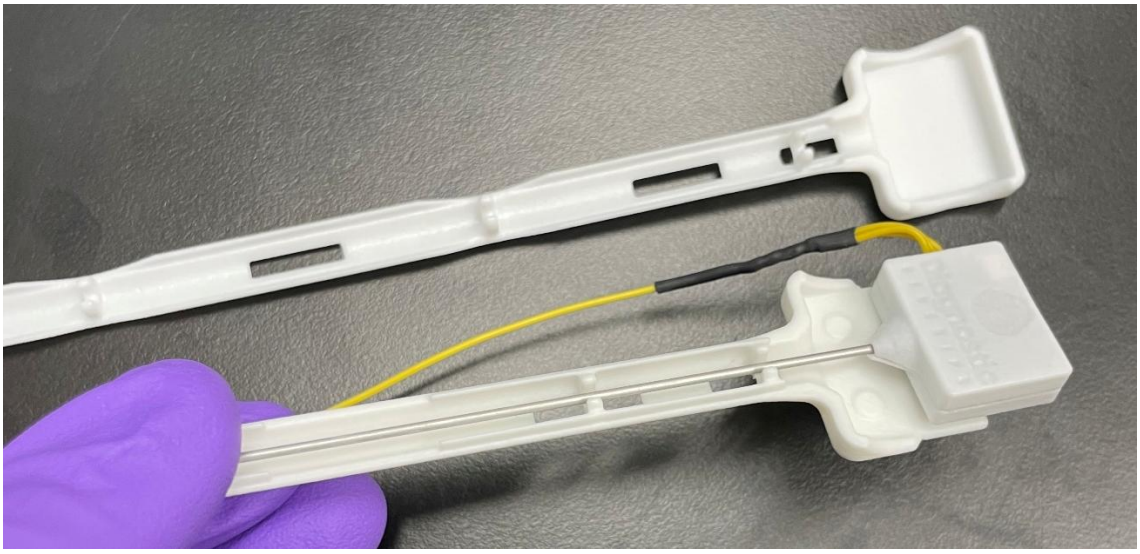
10. Remove the device from the sterile pouch





11. Remove the 4x clasps from the probe holder using your thumb as shown





12. Separate the top of the probe holder by sliding your thumb under the opening at the distal end up to the PCB backend.
    a. Use your thumb to stabilize the probe shank as you disassemble the top of the holder. This will prevent the probe shank from bending or the backend from lifting away with the top holder

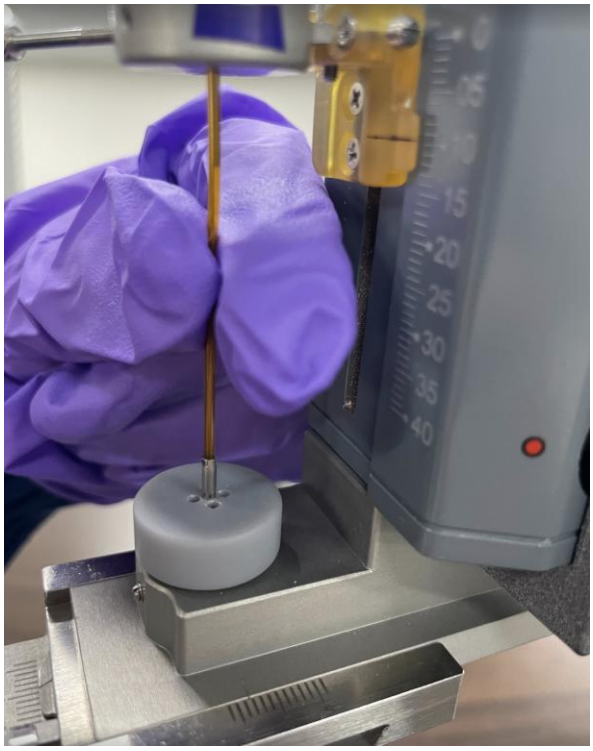
13. Now remove the probe from the bottom holder. You may discard the carrier components.



14. Verify that the polyimide skirt completely covers the distal end of the probe before proceeding.
    a. **WARNING** – attempting to insert the probe into a surgical drive without this protective skirt covering the probe tip may result in damage to the probe.



15. Carefully load the probe onto the surgical drive by guiding the polyimide skirt through the desired port. Ensure that the probe tip remains inside of the polyimide skirt during this step.

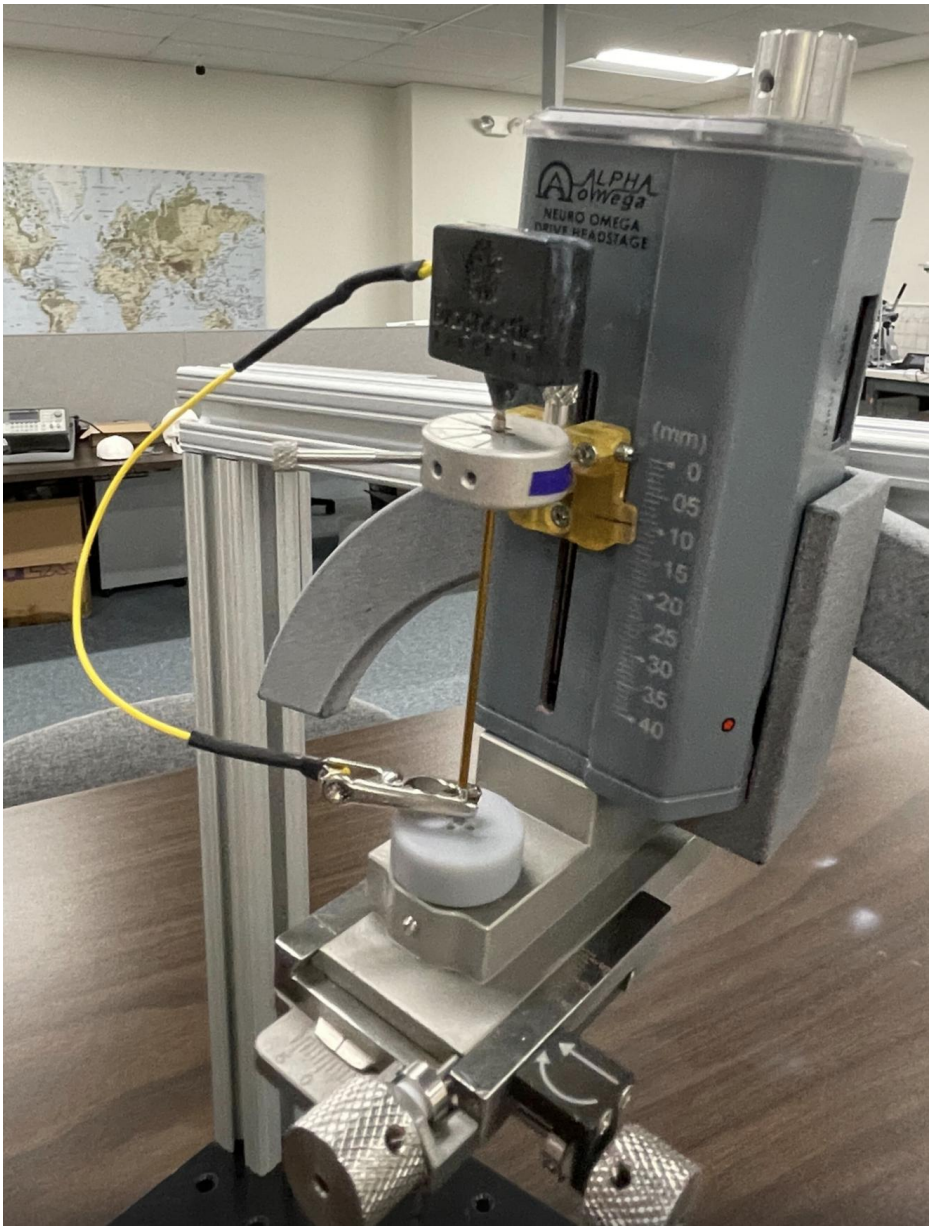




16. Continue to guide the polyimide skirt into the cannula while simultaneously lowering the probe further onto the drive.

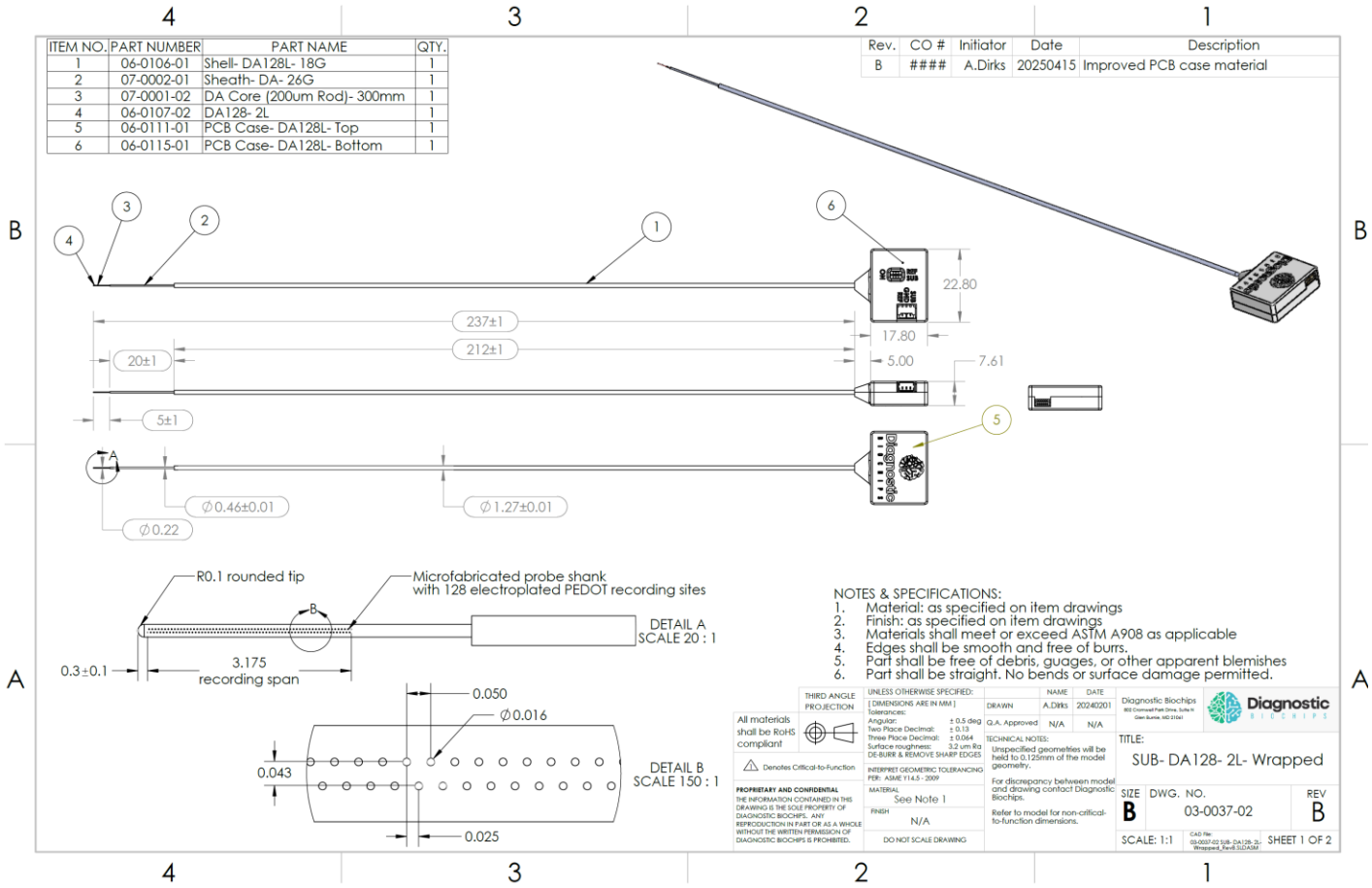


17. Lower the probe the rest of the way until it the backend contacts the metal collar on the polyimide skirt. The collar will act as a hard stop. The collar is 2mm long.
    a. Proceed to lock the probe in place using the set screw on the surgical drive.
    b. **WARNING** – take care not to overtighten the set screw as you may risk damaging the probe.



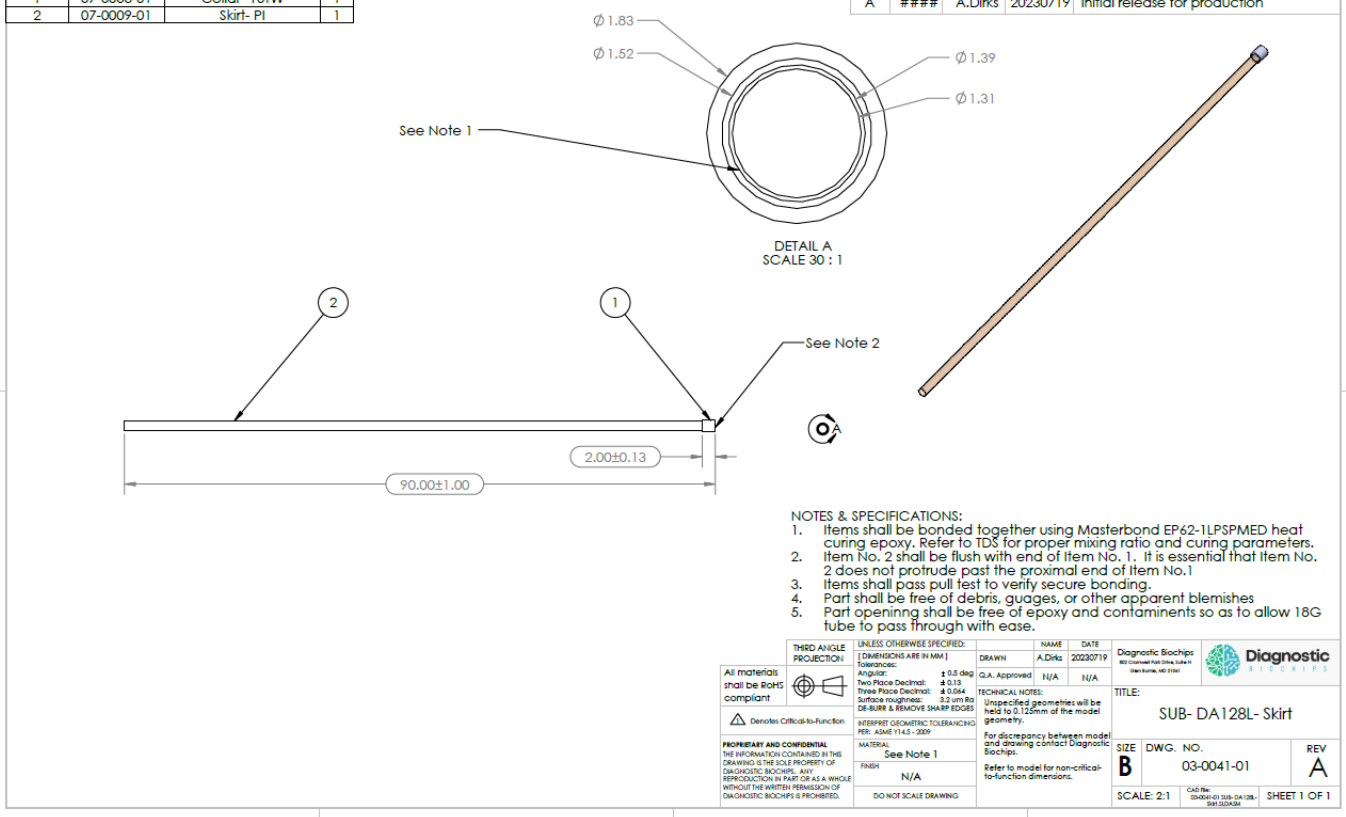
18. Using the alligator clip, secure the ground cable as desired.

6. ATTACHMENTS



ITEM NO.	PART NUMBER	PART NAME	QTY.
1	07-0008-01	Collar- 15TW	1
2	07-0009-01	Skirt- PI	1

Rev.	CO #	Initiator	Date	Description
A	####	A.Dirks	20230719	Initial release for production



NOTES & SPECIFICATIONS:
1. Items shall be bonded together using Masterbond EP62-1LPSPMED heat curing epoxy. Refer to TDS for proper mixing ratio and curing parameters.
2. Item No. 2 shall be flush with end of Item No. 1. It is essential that Item No. 2 does not protrude past the proximal end of Item No.1
3. Items shall pass pull test to verify secure bonding.
4. Part shall be free of debris, guages, or other apparent blemishes
5. Part openinng shall be free of epoxy and contaminents so as to allow 18G tube to pass through with ease.

THIRD ANGLE PROJECTION

All materials shall be RoHS compliant

Denotes Critical-to-Function

UNLESS OTHERWISE SPECIFIED: [DIMENSIONS ARE IN MM] Tolerances: Angular: ± 0.5 deg; Two Place Decimal: ± 0.13; Three Place Decimal: ± 0.064; Surface roughness: 3.2 um Ra; DE-BURR & REMOVE SHARP EDGES

INTERPRET GEOMETRIC TOLERANCING PER: ASME Y14.5 - 2009

MATERIAL: See Note 1

FINISH: N/A

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	A.Dirks	20230719
Q.A. Approved	N/A	N/A

TECHNICAL NOTES: Unspecified geometries will be held to 0.125mm of the model geometry.

For discrepancy between model and drawing contact Diagnostic Biochips.

Refer to model for non-critical-to-function dimensions.

PROPRIETARY AND CONFIDENTIAL
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Diagnostic Biochips
800 Cromwell Park Drive, Suite N
Glen Burnie, MD 21061

TITLE: SUB- DA128L- Skirt

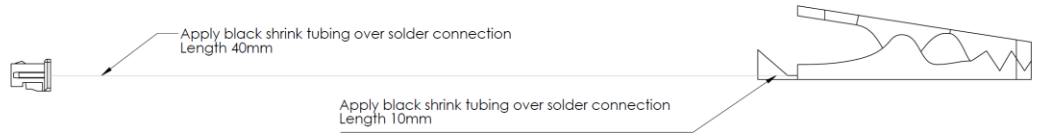
SIZE	DWG. NO.	REV
B	03-0041-01	A

SCALE: 2:1 | CAD file: 03-0041-01 SUB- DA128L- SKIRT.SLDASM | SHEET 1 OF 1

ITEM NO.	DBC P/N	Mfg P/N	Mfg Description	Mfg	QTY.
1	08-0009-01	51021-0300	Connector, 3 pin	Molex	1
2	08-0010-01	50058-8000	Contact, crimp	Molex	1
3	08-0011-01	314-1010-ND	BU-30, Gator clip, steel, non-insulated, 5A	Digikey	1

Rev.	CO #	Initiator	Date	Description
A	####	A.Dirks	20230719	Initial release





NOTES & SPECIFICATIONS:
1. All materials must be RoHS compliant.
2. Test 100% for continuity, opens, and shorts.

THIRD ANGLE PROJECTION

All materials shall be RoHS compliant

Denotes Critical-to-Function

UNLESS OTHERWISE SPECIFIED:
[DIMENSIONS ARE IN MM]
Tolerances:
Angular: ± 0.5 deg
Two Place Decimal: ± 0.13
Three Place Decimal: ± 0.064
Surface roughness: 3.2 um Ra
DE-BURR & REMOVE SHARP EDGES
INTERPRET GEOMETRIC TOLERANCING PER: ASME Y14.5 - 2009
MATERIAL: See Note 1
FINISH: N/A
DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	A.Dirks	20230719
Q.A. Approved	N/A	N/A

TECHNICAL NOTES:
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Diagnostic Biochips
800 Cromwell Park Drive, Suite N
Glen Burnie, MD 21061

TITLE: Cable Assembly- GND_Ref-DA128L

SIZE	DWG. NO.	REV
B	03-0044-01	A

SCALE: 3:1 | CAD File: | SHEET 1 OF 1

Diagnostic BIOCHIPS **Deep Array 128-2L**

REF DA128-2L

YYYY-MM-DD

LOT 0000X

YYYY-MM-DD

STERILE EO

CAUTION - INVESTIGATIONAL DEVICE
Limited by Federal (or United States) law to investigational use



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02-0010-01_A

7. REVISION HISTORY

Rev	Date	Author	Description
A	11/12/2024	A. Dirks	Initial release.